

Session 8 Quiz - Large Language Models

Track B | Grades 9–12 | 20 Questions

Q1 [Type: MCQ | Difficulty: Easy | QID :52802]

A type of AI trained on enormous amounts of text to generate human-like writing, answer questions, and translate languages is called a _____.

- A. Large Language Model
- B. Image Generator
- C. Recommendation System
- D. Search Engine

Correct Answer: A

Explanation: Large Language Models (LLMs) are trained on massive text datasets and generate language-based outputs. ChatGPT, Claude, and Gemini are examples. Image generators create images, not text. Recommendation systems suggest content. Search engines retrieve existing pages rather than generating new text.

Q2 [Type: MCQ | Difficulty: Easy | QID :]

Inside an LLM, words and parts of words are converted into small units called _____ before the model processes them.

- A. Pixels
- B. Files
- C. Tokens
- D. Folders

Correct Answer: C

Explanation: LLMs break text into tokens - small pieces that can be whole words, parts of words, or punctuation. The model reads and generates tokens, not raw letters. Pixels are units of images. Files and folders are storage concepts unrelated to language processing.

Q3 [Type: MCQ | Difficulty: Easy | QID :52815]

When a LLM produces a confident-sounding response that is actually false or made up, this is called a _____.

- A. Reflection
- B. Translation
- C. Prediction
- D. Hallucination

Correct Answer: D

Explanation: Hallucinations are when LLMs generate false information that sounds plausible. They happen because LLMs match patterns rather than retrieve verified facts. Reflection and translation are different operations. Prediction is what LLMs do at every step - but only some predictions become hallucinations.

Q4 [Type: MCQ | Difficulty: Easy | QID : 52817]

When you ask an LLM a factual question, it searches a database of verified facts to give you the correct answer.

- A. True
- B. False

Correct Answer: B (False)

Explanation: LLMs do not have a database of facts. They generate responses by predicting which tokens (words) are likely to come next based on patterns learned during training. This is why they can be confidently wrong - they are pattern-matching, not fact-checking. Treating an LLM like a database leads to trusting hallucinated answers.

Q5 [Type: MCQ | Difficulty: Medium | QID : 52819]

If you ask an LLM about an event that happened yesterday, you should trust its answer because it has up-to-date information about current events.

- A. True
- B. False

Correct Answer: B (False)

Explanation: LLMs have a knowledge cutoff - they were trained on data up to a specific date and have no information about anything that happened after. Asking about a recent event will often produce a confident-sounding answer based on outdated patterns, or a hallucinated guess. The model cannot access the internet in real time unless explicitly equipped with a search tool.

Q6 [Type: MCQ | Difficulty: Medium | QID : 52820]

LLMs sound human because they have been trained on enormous amounts of human-written text and learned its patterns.

- A. True
- B. False

Correct Answer: A (True)

Explanation: LLMs are "parrots" that imitate human fluency without any underlying mind or feelings. They learned from text written by humans who understood, so they reproduce that understanding-style fluency - but the model itself is pattern-matching, not thinking. Convincing imitation does not equal comprehension.

Q7 [Type: MCQ | Difficulty: Easy | QID : 52825]

What is a token in the context of LLMs?

- A. A type of cryptocurrency used to pay for accessing AI services online
- B. A reward the model receives for generating correct answers during training
- C. A security key required to access the model and use its features
- D. A small unit of text - a word, part of a word, or punctuation

Correct Answer: D

Explanation: LLMs break text into tokens for processing. A single token might be a whole word like "cat," a word fragment like "ed" or "'s," or a punctuation mark. This is why a sentence often has more tokens than words. The other options describe completely unrelated concepts that share the word "token" in other contexts.

Q8 [Type: MCQ | Difficulty: Medium | QID : 52827]

What does an LLM's context window define?

- A. The total number of parameters the model has learned during training
- B. How creative or random the model's outputs are when generating responses
- C. The maximum amount of text the model can "see" at once
- D. The number of different languages the model can understand and speak fluently

Correct Answer: C

Explanation: Context window = the model's short-term memory limit, like a whiteboard with finite space. When the whiteboard fills up, earlier text gets pushed out. This is why long conversations eventually "forget" what was said at the beginning. The other options describe other LLM properties that are not the context window.

Q9 [Type: MCQ | Difficulty: Medium | QID :52830]

Parameters in an LLM are best described as:

- A.** Internal numerical values that store the patterns the model learned during training
- B.** The exact sentences and phrases the model has memorised word-for-word from its training data
- C.** The instructions or prompts that a user types into the chat to get a response
- D.** Fixed rules and logic written by programmers to control the model's behaviour

Correct Answer: A

Explanation: Parameters are the "brain cells" or memory knobs of the model - internal numbers adjusted during training to store learned patterns. GPT-3 has 175 billion of these. Option B confuses parameters with stored text. Option C describes prompts. Option D describes traditional programming, which LLMs are not.

Q10 [Type: MCQ | Difficulty: Medium | QID :52832]

Which mechanism acts like a "spotlight" to help the model focus on the most relevant words in a long sentence?

- A.** Tokens
- B.** Parameters
- C.** Context window
- D.** Attention

Correct Answer: D

Explanation: Attention allows the model to weigh the importance of different words and connect them across long distances - for example, linking "cat" near the start of a sentence to "jumped" near the end, even when many words separate them. Before attention, AI models would "forget" the beginning of a long sentence by the time they reached the end. The other options are real LLM concepts but serve different functions.

Q11 [Type: MCQ | Difficulty: Medium | QID : 52834]

What does the "parrot analogy" explain about Large Language Models?

- A.** LLMs are brightly coloured creatures that are easily distracted by their surroundings
- B.** LLMs can mimic human speech without truly understanding the meaning
- C.** LLMs have a better memory for spoken words than they do for written text on a page
- D.** LLMs are capable of learning to speak fluently by listening to their owners over time

Correct Answer: B

Explanation: The parrot analogy highlights that imitation of language does not equal the existence of a mind or understanding. A parrot can say "Hello" at exactly the right moment - but it has no idea what "Hello" means. LLMs work the same way: incredibly convincing imitation, no inner experience. The other options invent unrelated parrot traits.

Q12 [Type: MCQ | Difficulty: Medium | QID : 52840]

Which of the following is a symptom of an LLM lacking "common sense"?

- A.** It forgets the user's name after processing about 100,000 words of conversation
- B.** It refuses to answer any questions about the solar system or space exploration
- C.** It can explain complex physics but says you can fry an egg on the sun
- D.** It can only communicate in binary code

Correct Answer: C

Explanation: LLMs often fail at basic logical reasoning that humans find obvious, even when they can explain scientific facts at length. The model can produce a perfectly accurate explanation of solar temperature, then immediately suggest doing impossible things on the sun - because pattern matching does not include real-world common sense. Option A describes the context window. Options B and D are fabricated behaviours.

Q13 [Type: MCQ | Difficulty: Medium | QID : 52850]

Which of the following best illustrates how LLMs work - even though the prompts sound very different?

- A. Detecting whether a photo contains a cat by analysing its visual features
- B. Writing a haiku and translating "hello" - both reduce to next-token prediction
- C. Counting the number of pixels in an image to measure its resolution
- D. Encrypting a file with a secure algorithm before sending it over the Internet

Correct Answer: B

Explanation: Diverse-looking tasks share the same underlying mechanism. Whether the user asks for a haiku, a translation, a code snippet, or an explanation of photosynthesis - the model is always doing the same thing, predicting the next token one at a time. Options B and C describe image/discriminative tasks. Option D is unrelated.

Q14 [Type: MCQ | Difficulty: Hard | QID :]

A friend says: "Bigger LLMs (with more parameters) are ALWAYS better than smaller ones." What is the most balanced response?

- A. Correct - parameter count is the only thing that matters when comparing LLMs
- B. Smaller models are always better than larger ones because they run faster and cheaper
- C. Parameters do not affect model performance at all and can be safely ignored
- D. Bigger isn't always better - a well-built smaller model can outperform larger ones

Correct Answer: D

Explanation: Size matters but is not everything. Larger models have more capacity to learn complex patterns, but they are also expensive to train and run, and a smaller model with high-quality training data can often beat a larger one. Smaller models are also faster, cheaper, and can run on phones. Options A, B, and C each represent an extreme position

Q15 [Type: MCQ | Difficulty: Hard | QID :]

Why might an LLM associate certain jobs with specific genders in its responses?

- A.** Because it is specifically programmed to have its own opinions about gender roles
- B.** Because it has directly observed humans in the real world during its training process
- C.** Because it repeats stereotypes and biases present in its training data
- D.** Because it uses logical reasoning to determine which jobs suit which genders best

Correct Answer: C

Explanation: If the human-written training text contains gender stereotypes - for example, doctors referred to as "he" and nurses as "she" - the model learns those statistical patterns and replicates them in its outputs. The model has no opinions (A), cannot observe the real world (B), and is not reasoning logically about job suitability (D) - it is mirroring patterns from its training corpus, which is the same GIGO principle

Q16 [Type: MCQ | Difficulty: Hard | QID : 52857]

A principal asks: "What is the difference between using a calculator on math homework and using an LLM on essay homework?" Which best captures the distinction?

- A.** There is no meaningful difference - both are tools that help students complete their schoolwork faster
- B.** Calculators are older technology, which makes them more socially acceptable in classrooms today
- C.** LLMs are far more accurate than calculators, so they are actually more appropriate for homework
- D.** A calculator handles one step; an LLM can do the entire task being assessed

Correct Answer: D

Explanation: A calculator performs a defined operation the student already understands within a larger problem they are solving. An LLM can perform the entire cognitive task - the thinking the assignment was designed to develop. That is the key distinction. Option A misses the asymmetry. Option B confuses technology age with appropriateness. Option C is factually questionable and misses the point.

Q17 [Type: MCQ Scenario | Difficulty: Medium | QID : 52859]

Scenario: A user begins a 200-message chat with an LLM. By message 180, the model has clearly forgotten the user's name mentioned in message 2.

Question: Which concept BEST explains this?

- A.** The context window has filled up, pushing earlier text out of memory
- B.** The model is overfit to the user's name
- C.** The training data did not include long conversations
- D.** The model needs more parameters to remember

Correct Answer: A

Explanation: The context window holds a fixed number of tokens, and when the whiteboard fills up, the oldest content drops off and the model can no longer see it. Option B misuses an unrelated term. Options C and D blame the wrong components of LLM architecture.

Q18 [Type: MCQ Scenario | Difficulty: Hard | QID :52861]

Scenario: A medical chatbot built on an LLM tells a user their symptoms suggest a rare disease - and confidently gives detailed treatment advice.

Question: Of the following LLM limitations, which combination most directly creates risk in this scenario?

- A.** Limited context window plus too many tokens
- B.** No true understanding, cannot verify facts, and may produce harmful output without safeguards
- C.** Big parameter count plus small attention window
- D.** Outdated training data plus small context window

Correct Answer: B

Explanation: The LLM lacks understanding of medicine, has no internal fact-checker, and can reproduce biased or harmful content without realising it is doing so. The other options pick limitations that exist but do not best capture the danger here - the issue is not the size of the context window or the number of parameters, but the absence of medical understanding combined with confident-sounding output.

Q19 [Type: FRQ - Short Answer | Difficulty: Hard | QID : 52863]

Scenario: A patient researches their symptoms online and finds that an LLM gives a detailed, confident-sounding response describing a possible diagnosis.

Question: Why should they not act on this information without consulting a medical professional?

(Respond in 3–4 sentences.)

Model Answer: LLMs generate responses by predicting likely word sequences from training data, not by actually examining the patient or diagnosing medical conditions - and they can sound confident even when wrong. A medical diagnosis requires examination, medical history review, and clinical judgment the LLM cannot provide. Acting on a hallucinated answer could lead the patient to ignore serious symptoms or self-treat for a condition they do not have.

Q20 [Type: FRQ - Scenario Analysis | Difficulty: Hard | QID : 52865]

Scenario: A district court judge is considering whether to allow lawyers in their courtroom to use LLMs to help draft legal filings. Some lawyers argue it will save time and reduce client costs. The judge is concerned about reliability.

Question: Identify one significant benefit AND one significant risk of allowing LLM-assisted filings, and describe what the court should require to balance them.

(Respond in 3–4 sentences.)

Model Answer: A significant benefit is that LLMs can help lawyers draft routine filings faster, reducing legal costs and making representation more affordable. A significant risk is that LLMs can hallucinate case citations, misrepresent precedents, or produce filings with factual errors - there have been real cases where lawyers submitted fake citations from LLMs to courts. To balance these, the court should require that any lawyer using LLM assistance verify all factual claims and citations independently, disclose LLM use in filings, and accept full professional responsibility for the content submitted.